Lab Report No 6

Digital Signal Processing

Submitted By: AKMAL SULTAN

Registration No: 23PWCSE2253

Section: B

Department of Computer Systems Engineering

University of Engineering and Technology Peshawar

CSE 402L: Digital Signal Processing

Demonstration of Concepts	**Poor (Does not meet expectation (1))** The student failed to demonstrate a clear understanding of the assignment concepts	**Fair (Meet Expectation (2-3))** The student demonstrated a clear understanding of some of the assignment concepts	**Good (Exceeds Expectation (4-5)** The student demonstrated a clear understanding of the assignment concepts	**Score** **30%**
Accuracy	The student completed (<50%) tasks and provided MATLAB code and/or Simulink models with errors. Outputs shown are not correct in form of graphs (no labels) and/or tables along with incorrect analysis or remarks.	The student completed partial tasks (50% - <90%) with accurate MATLAB code and/or Simulink models. Correct outputs are shown in form of graphs (without labels) and/or tables along with correct analysis or remarks.	The student completed all required tasks (90%-100%) with accurate MATLAB code and/or Simulink models. Correct outputs are shown in form of labeled graphs and/or tables along with correct analysis or remarks.	**30%**
Following Directions	The student clearly failed to follow the verbal and written instructions to successfully complete the lab	The student failed to follow the some of the verbal and written instructions to successfully complete all requirements of the lab	The student followed the verbal and written instructions to successfully complete requirements of the lab	**20%**
Time Utilization	The student failed to complete even part of the lab in the allotted amount of time	The student failed to complete the entire lab in the allotted amount of time	The student completed the lab in its entirety in the allotted amount of time	**20%**

Lab 6: Signal Processing Training

Title: Signal processing onramp

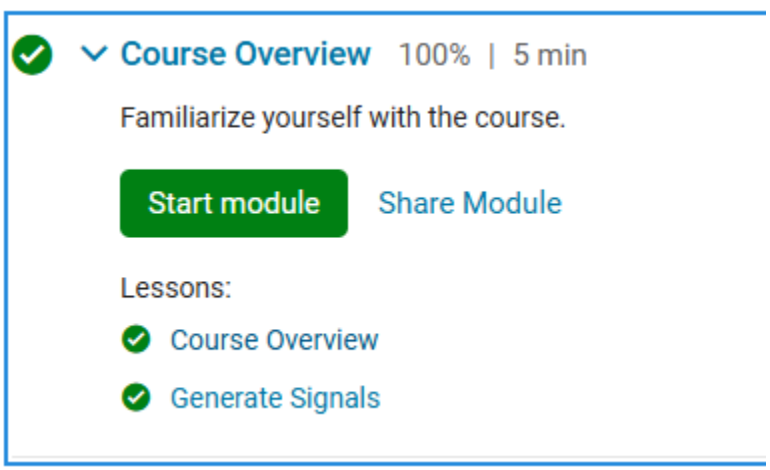
Learn basics of practical signal processing techniques in MATLAB. Use spectral analysis and filtering techniques to process, analyze, and extract information form signal data. Visit the following website: https://matlabacademy.mathworks.com/details/signal-processing-onramp/signalprocessing and perform the following tasks and attach the Certificate/ Progress Report acquired from MathWorks as part of the lab Report

Objectives

1. **Course Overview**
    a. Familiarize yourself with the course.
    b. Remarks along with final snapshot.
2. **Spectral Analysis Workflow**
    a. Import Signals into MATLAB and view power spectra.
    b. Remarks along with final snapshot.
3. **Preprocessing Signals**
    a. Clean up time base and align signals.
    b. Remarks along with final snapshot.
4. **Spectral Analysis**
    a. Perform spectral analysis to view signals in the frequency domain.
    b. Remarks along with final snapshot.
5. **Filtering**
    a. Filter signals using basic techniques.
    b. Remarks along with final snapshot.
6. **Signal Measurements**
    a. Extract information from signals.
    b. Remarks along with final snapshot.
7. **Conclusion**
    a. Learn next steps and give feedback on the course.
    b. Remarks along with final snapshot.

1. **Course Overview**
    a. Familiarize yourself with the course.
    b. Remarks along with final snapshot.

- **Remarks and Final Snapshot:**
  The course was a fantastic introduction to key signal processing concepts. It has made learning interactive and practical, helping me quickly learn the concepts.



2. **Spectral Analysis Workflow**
    c. Import Signals into MATLAB and view power spectra.
    d. Remarks along with final snapshot.

Import a Signal

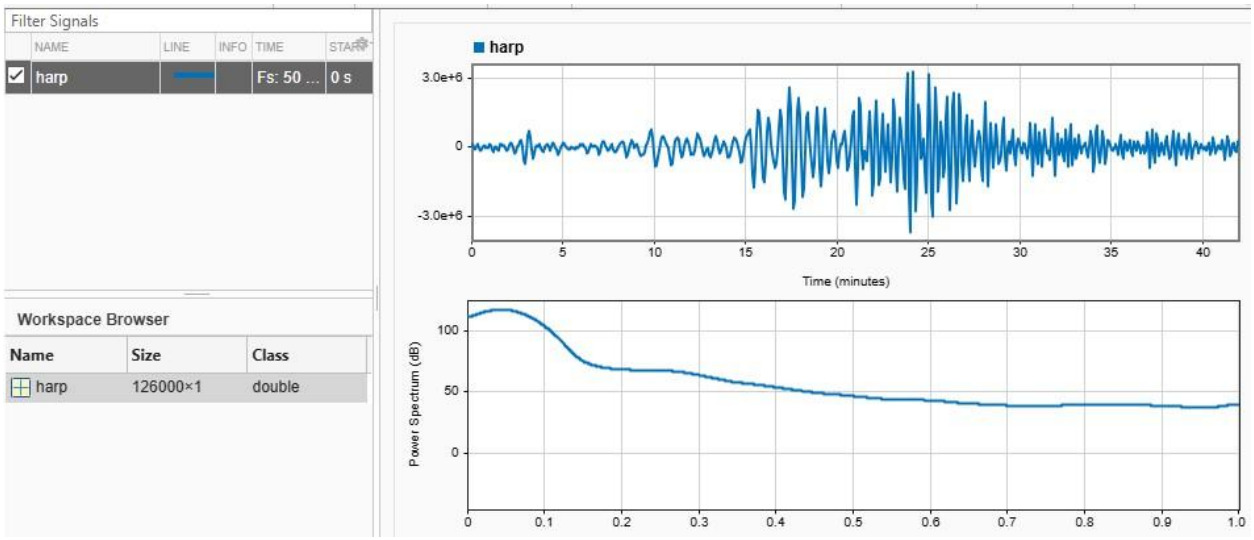
Instructions are in the task pane to the left. Complete and submit each task one at a time.

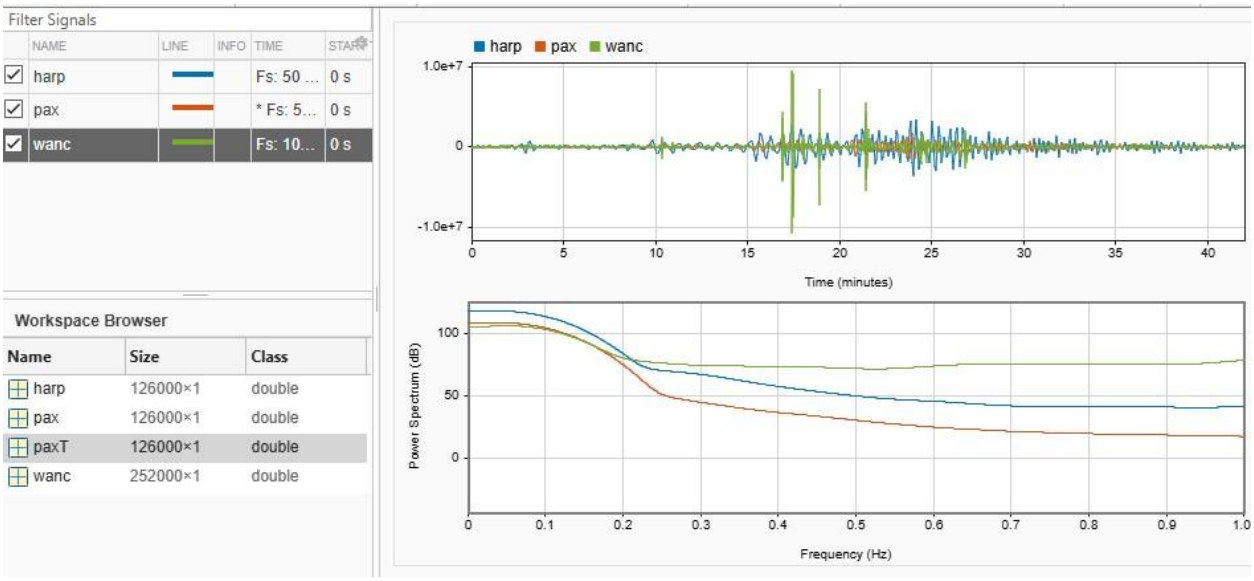
Task 1

```
harp = readmatrix("harp.csv")
plot(harp)
```

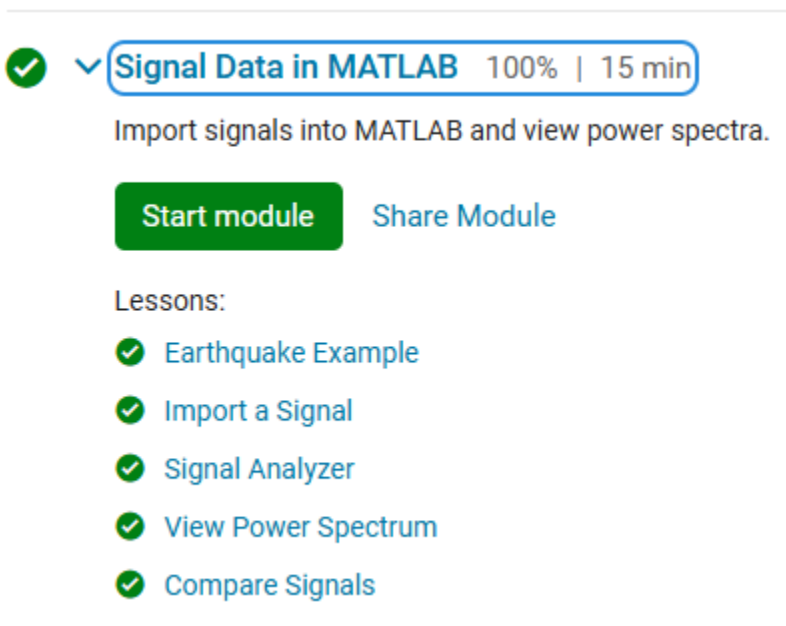
Further Practice

```
66141
66453
66758
67067
67371
67694
68001
68310
68651
69008
69384
69749
```

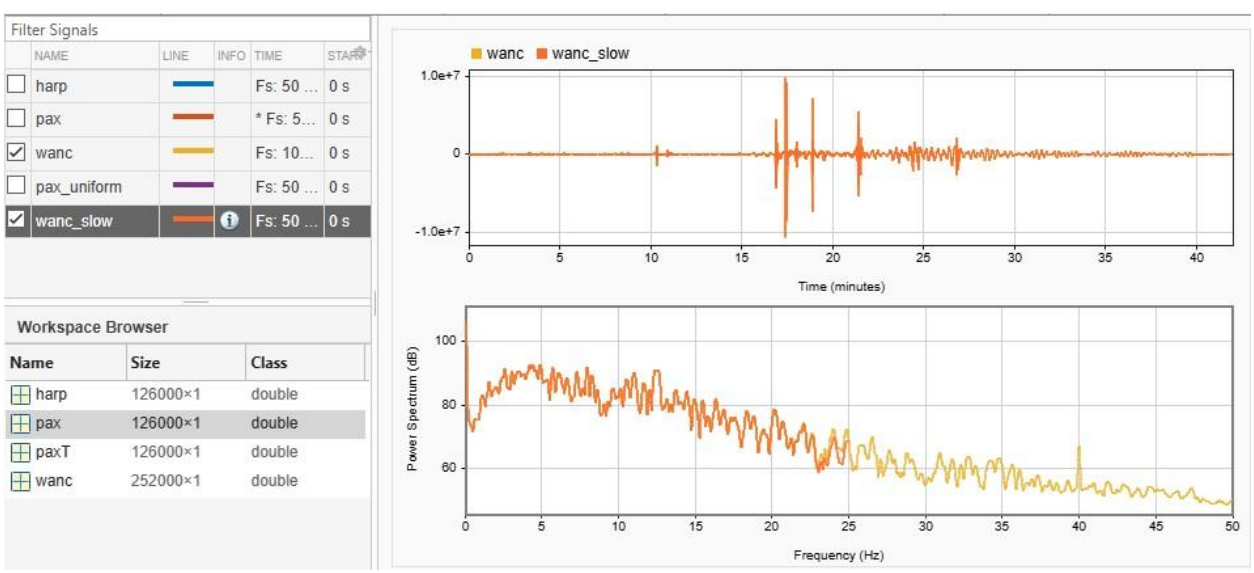


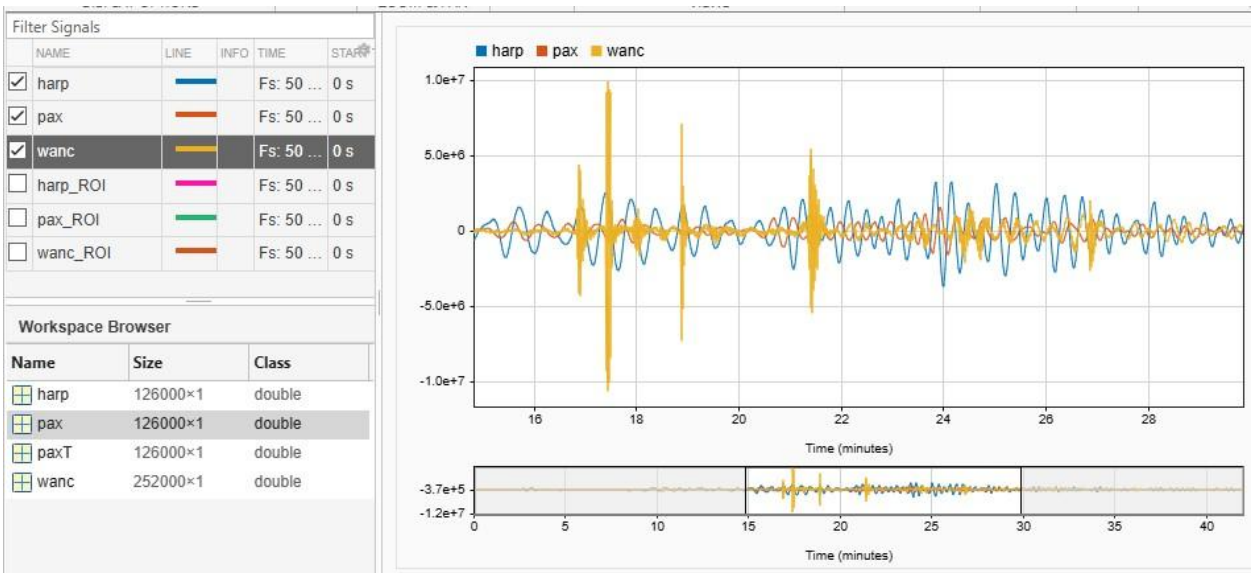


- **Remarks and Final Snapshot:**
  It provided a clear understanding of importing signals into MATLAB and analyzing their power spectra.

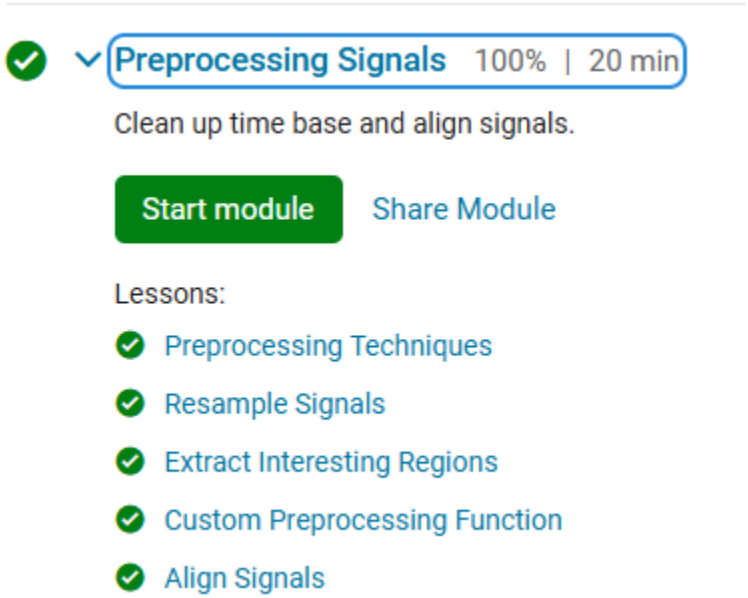


3. **Preprocessing Signals**
   a. Clean up time base and align signals.
   b. Remarks along with final snapshot





- **Remarks and Final Snapshot:**
  This section was essential for learning how to clean up the time base and align signals in MATLAB.



4. **Spectral Analysis:**
   a. Perform spectral analysis to view signals in the frequency domain.
   b. Remarks along with final snapshot.

Task 1

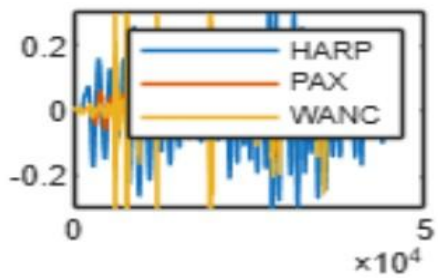
```
[harp,wanc] = alignsignals(harp,wanc);
```

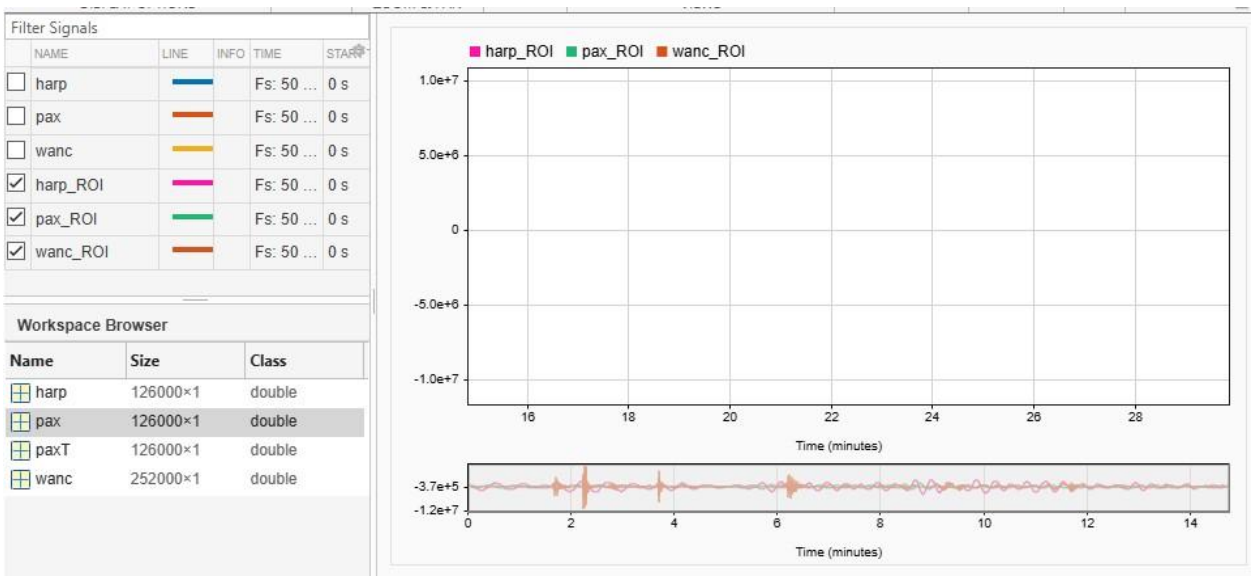
Task 2

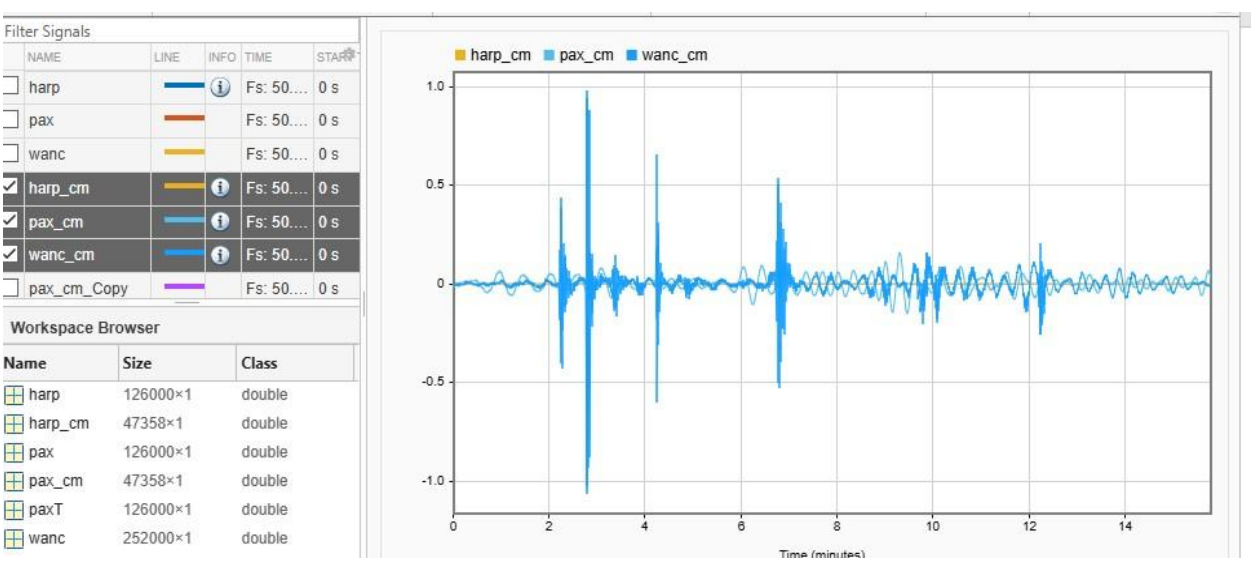
```
[pax,wanc] = alignsignals(pax,wanc);
```

This code plots the signals from the HARP and PAX stations.

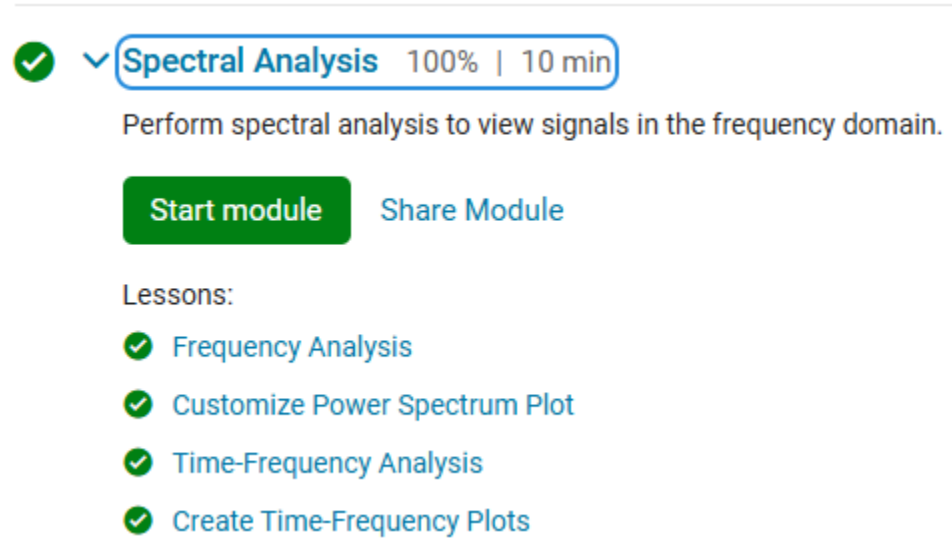
```
plot(harp)
```



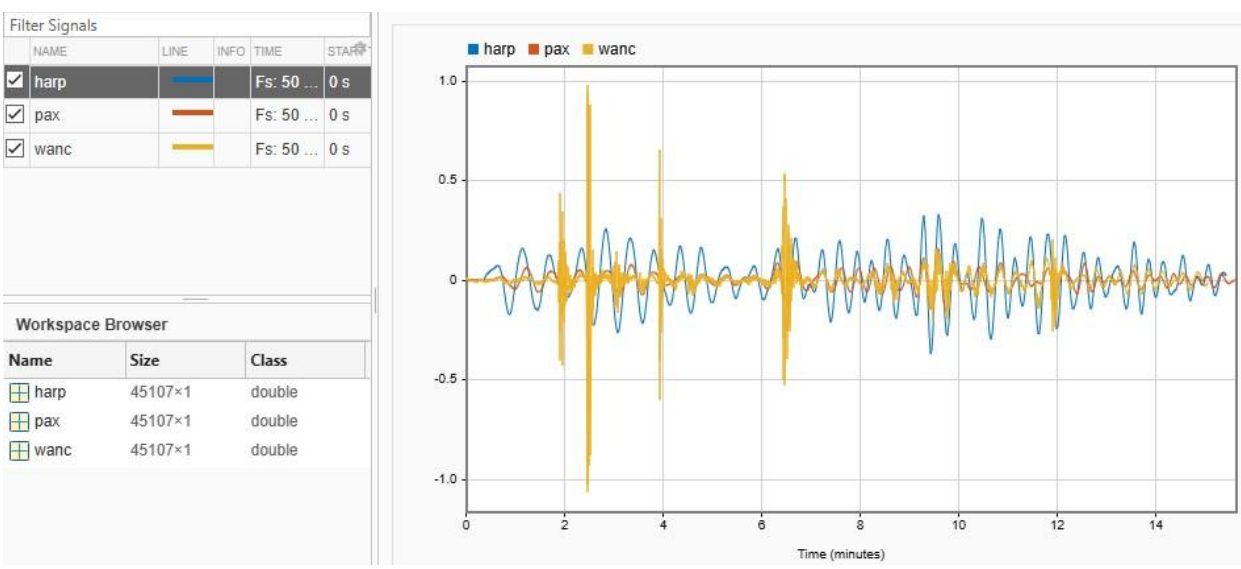


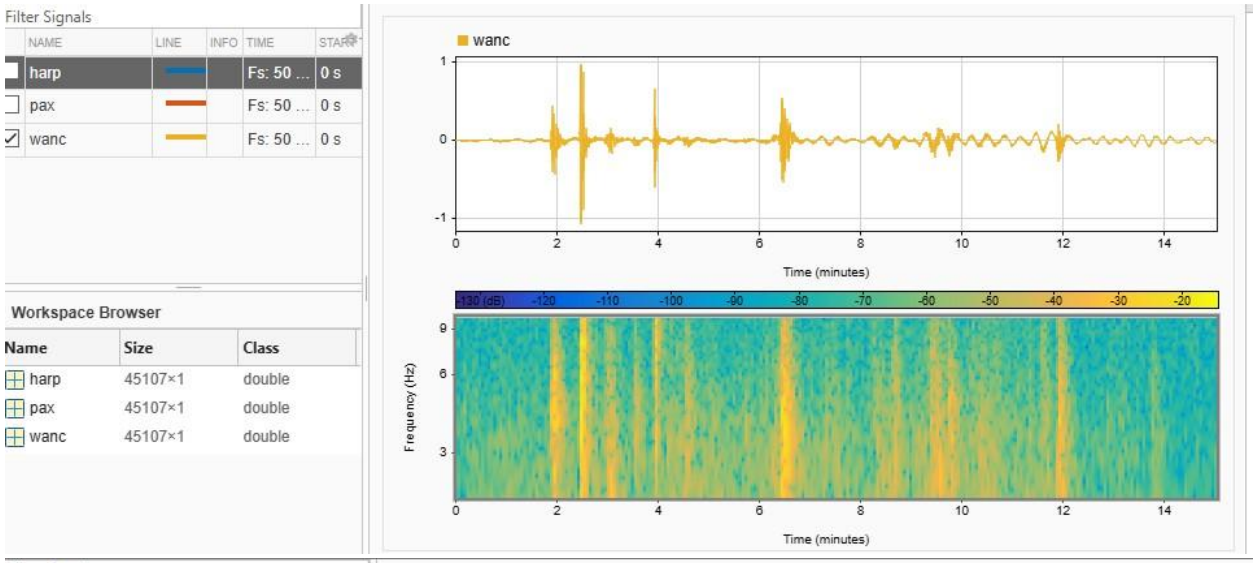


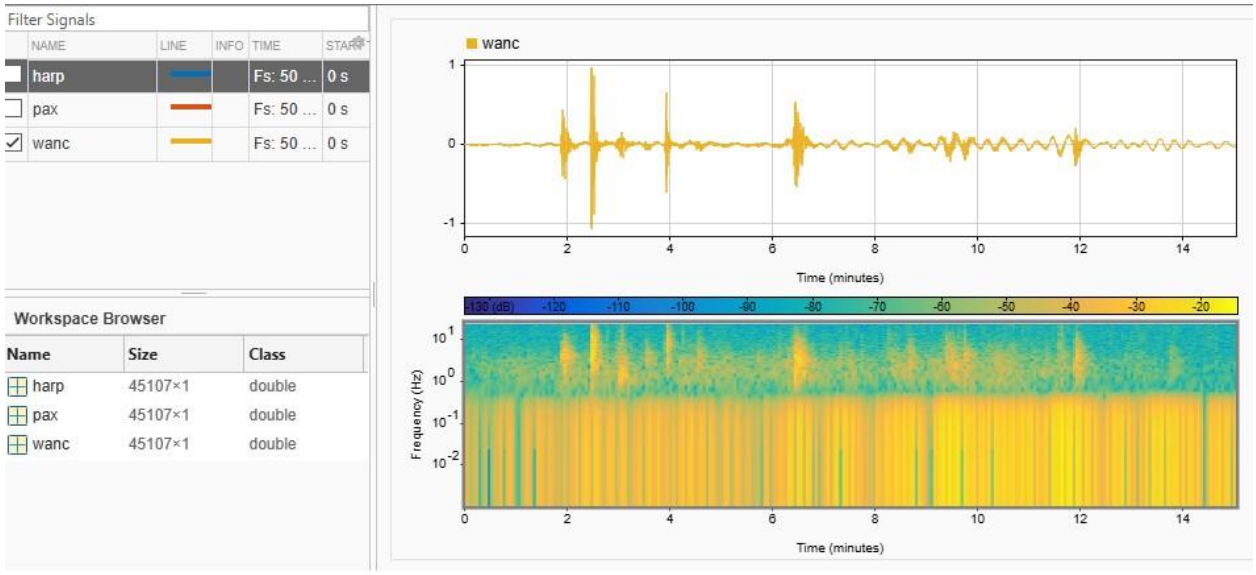
- **Remarks and Final Snapshot:**
  In this section we learned to view signals in the frequency domain using MATLAB. It helped me understand how to perform spectral analysis to identify frequency components

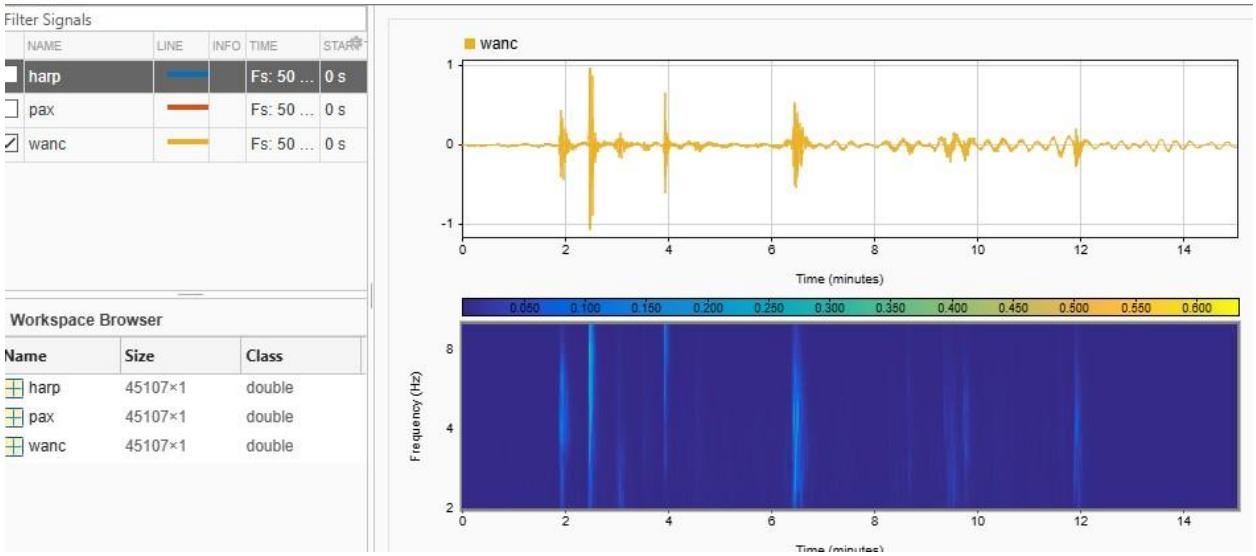


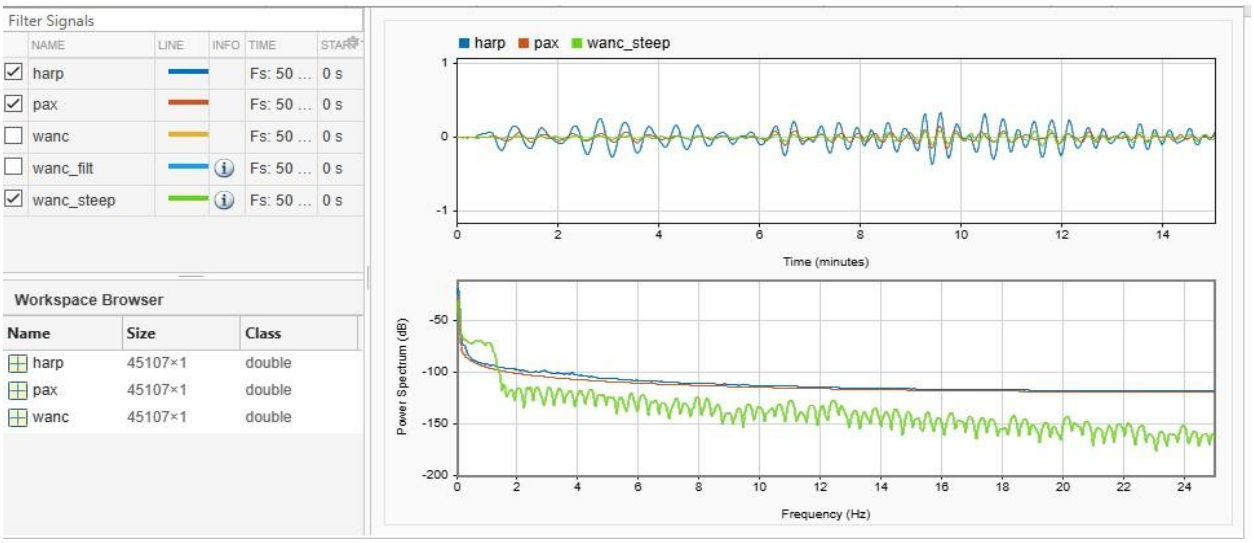
5. **Filtering**
   a. Filter signals using basic techniques.
   b. Remarks along with final snapshot.

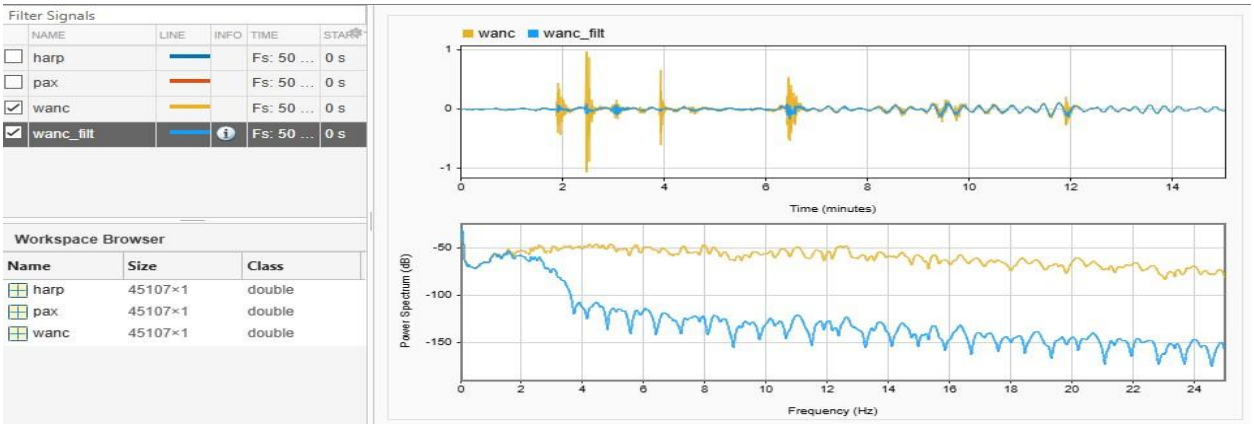


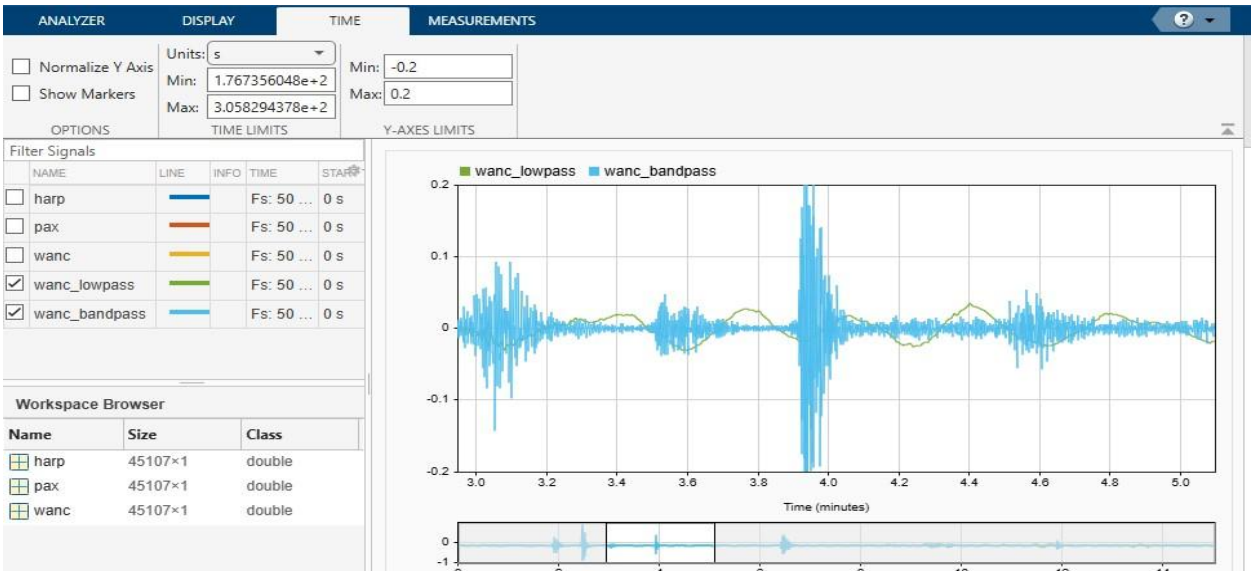

Filter Signals
NAME LINE INFO TIME START
harp Fs: 50 ... 0 s
pax Fs: 50 ... 0 s
wanc Fs: 50 ... 0 s
Workspace Browser
Name Size Class
harp 45107×1 double
pax 45107×1 double
wanc 45107×1 double
wanc
Time (minutes)
Frequency (Hz)
Time (minutes)


Filter Signals
NAME LINE INFO TIME START
harp Fs: 50 ... 0 s
pax Fs: 50 ... 0 s
wanc Fs: 50 ... 0 s
Workspace Browser
Name Size Class
harp 45107×1 double
pax 45107×1 double
wanc 45107×1 double
wanc
Time (minutes)
Frequency (Hz)
Time (minutes)

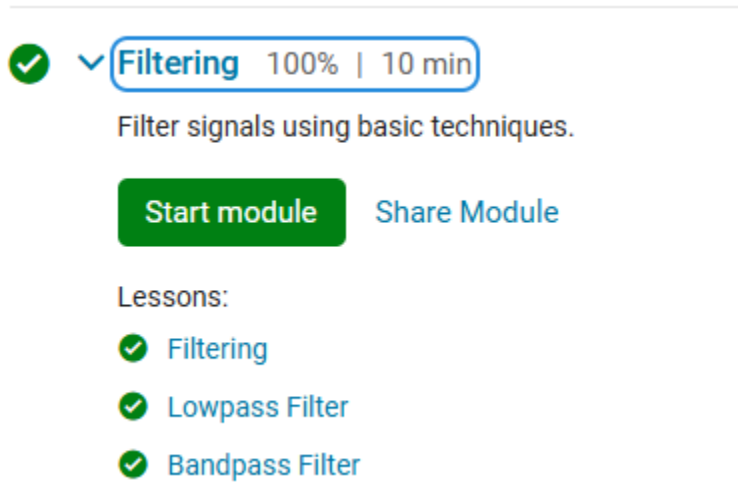

Filter Signals
NAME LINE INFO TIME START
harp Fs: 50 ... 0 s
pax Fs: 50 ... 0 s
wanc Fs: 50 ... 0 s
Workspace Browser
Name Size Class
harp 45107×1 double
pax 45107×1 double
wanc 45107×1 double
wanc
Time (minutes)
Frequency (Hz)


Filter Signals
NAME LINE INFO TIME START
harp Fs: 50 ... 0 s
pax Fs: 50 ... 0 s
wanc Fs: 50 ... 0 s
wanc_filt Fs: 50 ... 0 s
wanc_steep Fs: 50 ... 0 s
Workspace Browser
Name Size Class
harp 45107×1 double
pax 45107×1 double
wanc 45107×1 double
harp pax wanc_steep
Time (minutes)
Power Spectrum (dB)
Frequency (Hz)

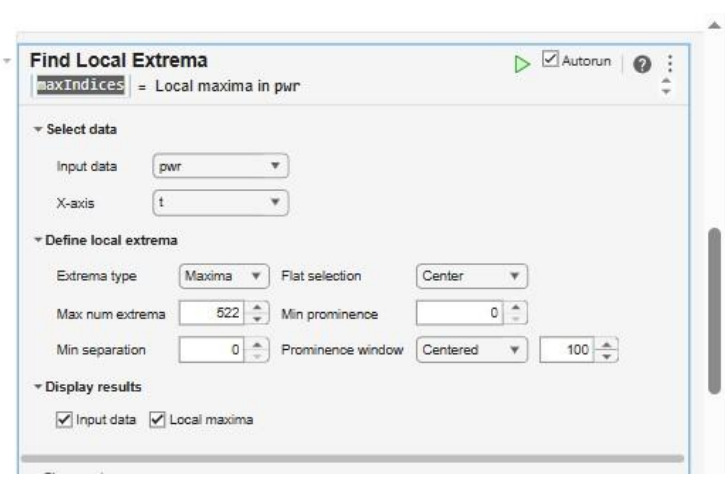

Filter Signals
NAME LINE INFO TIME START
harp Fs: 50 ... 0 s
pax Fs: 50 ... 0 s
wanc Fs: 50 ... 0 s
wanc_filt Fs: 50 ... 0 s
Workspace Browser
Name Size Class
harp 45107×1 double
pax 45107×1 double
wanc 45107×1 double
wanc wanc_filt
Time (minutes)
Power Spectrum (dB)
Frequency (Hz)

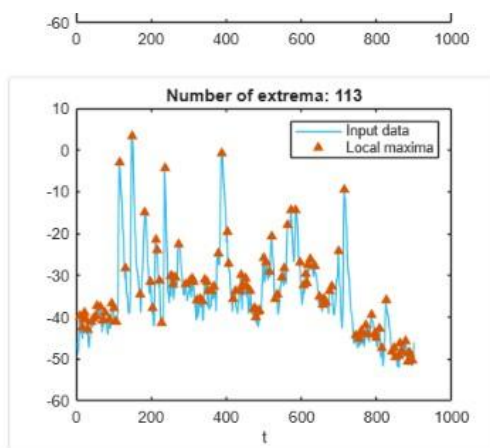

ANALYZER DISPLAY TIME MEASUREMENTS
Normalize Y Axis
Show Markers
OPTIONS
Units: s
Min: 1.767356048e+2
Max: 3.058294378e+2
TIME LIMITS
Min: -0.2
Max: 0.2
Y-AXES LIMITS
Filter Signals
NAME LINE INFO TIME START
harp Fs: 50 ... 0 s
pax Fs: 50 ... 0 s
wanc Fs: 50 ... 0 s
wanc_lowpass Fs: 50 ... 0 s
wanc_bandpass Fs: 50 ... 0 s
Workspace Browser
Name Size Class
harp 45107×1 double
pax 45107×1 double
wanc 45107×1 double
wanc_lowpass wanc_bandpass
Time (minutes)

- **Remarks and Final Snapshot:**
  In this section we learned basic signal filtering techniques in MATLAB. It provide practical experience to understand how to remove noise and isolate specific signal components.

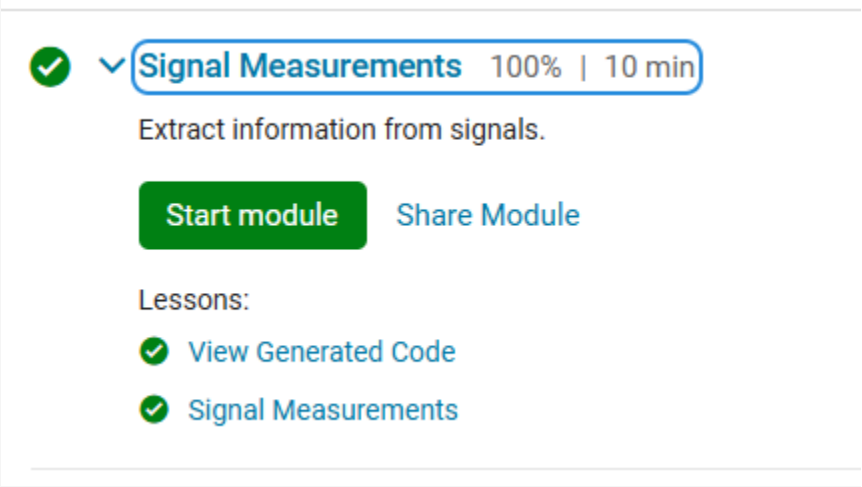


6. **Signal Measurements**
   a. Extract information from signals.
   b. Remarks along with final snapshot.

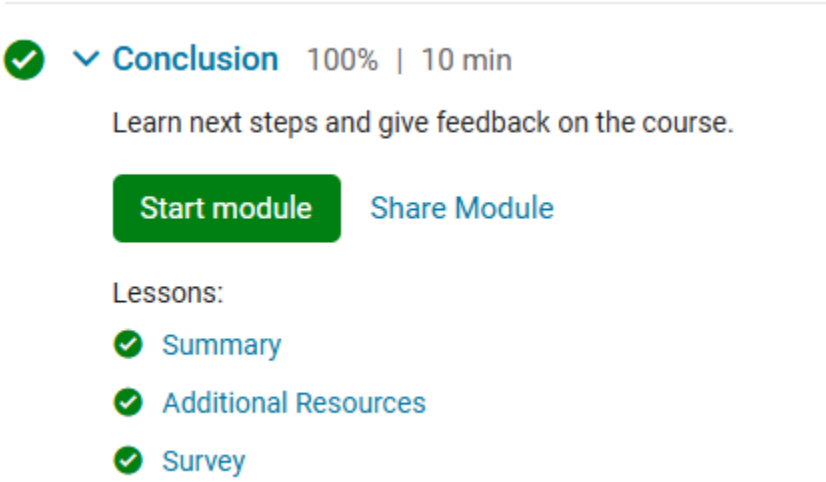




- **Remarks and Final Snapshot:**
  This part teaches how to extract valuable information from signals using MATLAB. It provided practical skills for analyzing signal characteristics.



7. **Conclusion**
   a. Learn next steps and give feedback on the course.
   b. Remarks along with final snapshot.



Course Completion Certificate:

MathWorks | Training Services

Course Completion Certificate

AKMAL SULTAN

has successfully completed **100%** of the self-paced training course

Signal Processing Onramp

DIRECTOR, TRAINING SERVICES

3 December 2025